Imperial College
London

Lecture 7

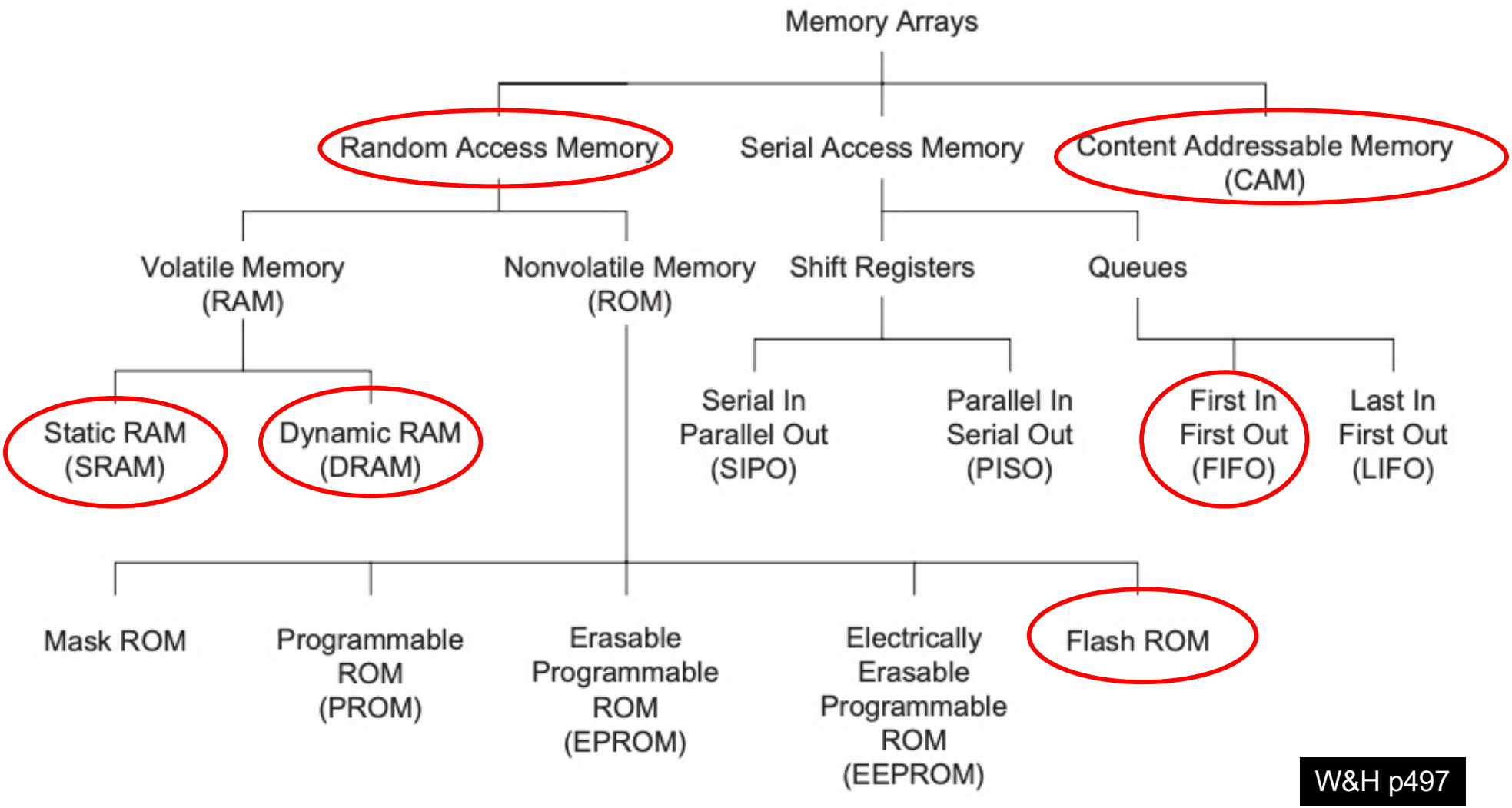
Array Subsystems

Prof Peter YK Cheung

Department of Electrical & Electronic Engineering

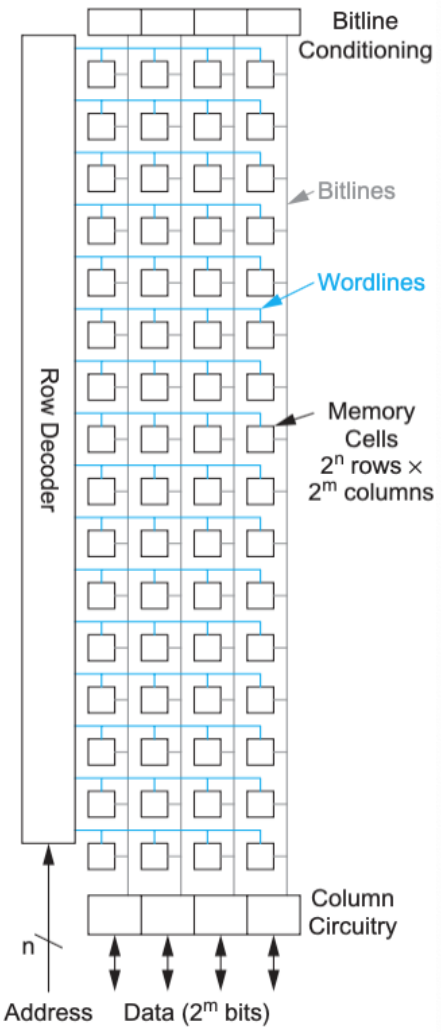
URL: https://elec70142-digital-vlsi-design.github.io/course-website/
E-mail: p.cheung@imperial.ac.uk

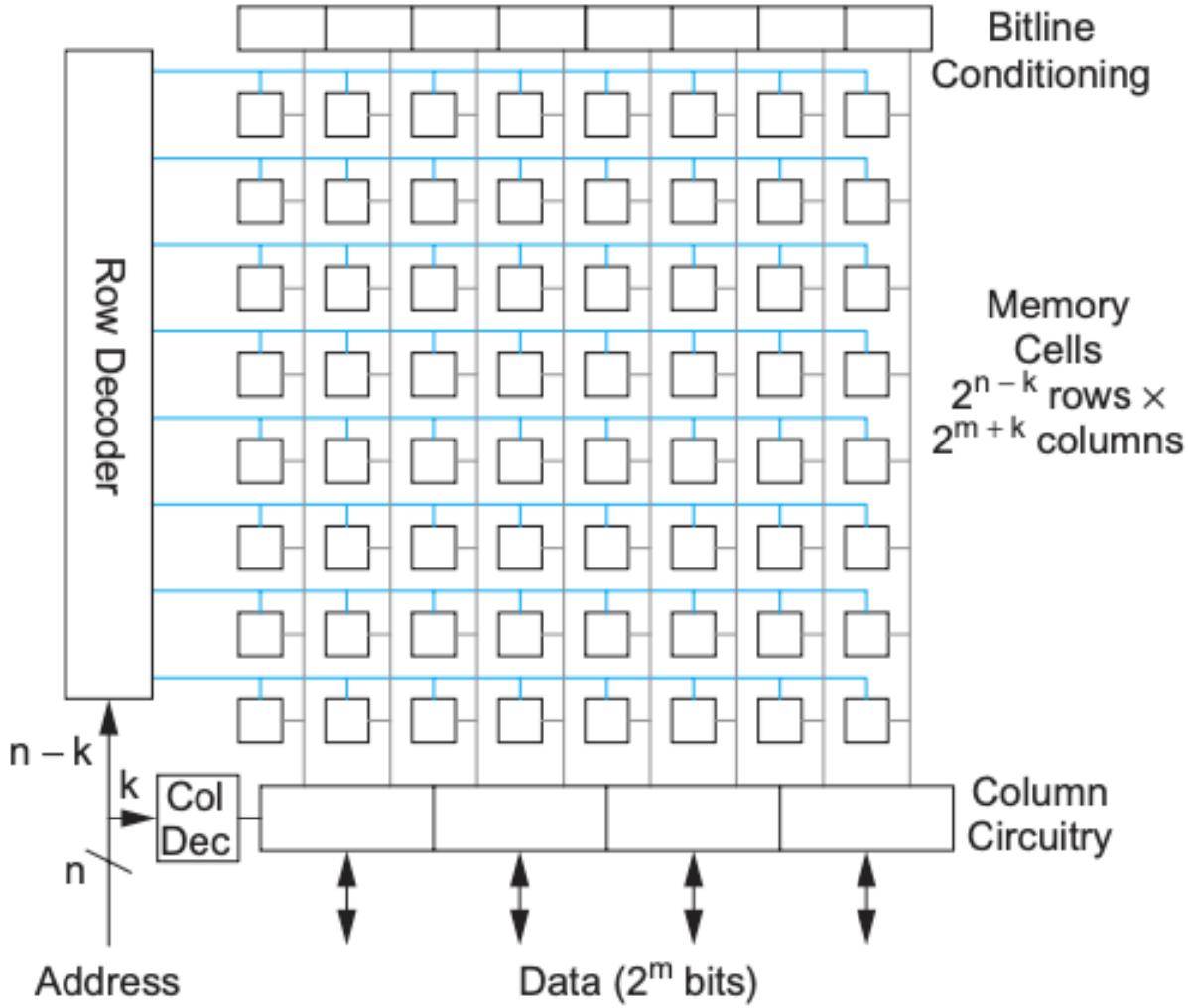
Categories of Memory Arrays



W&H p497

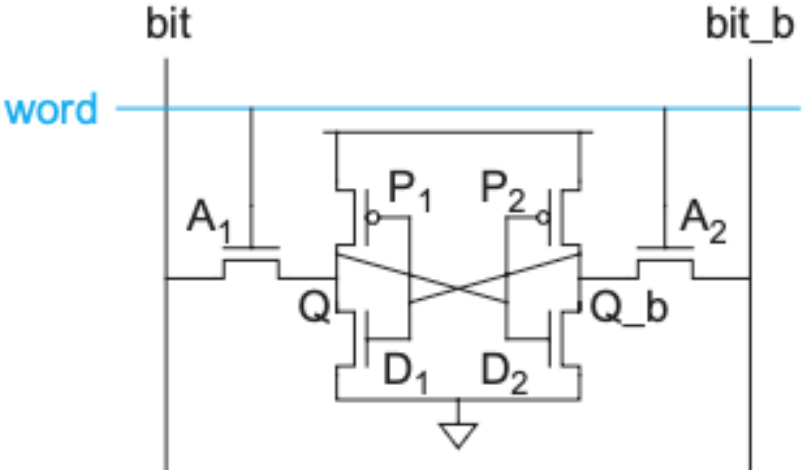
Memory Array Architecture

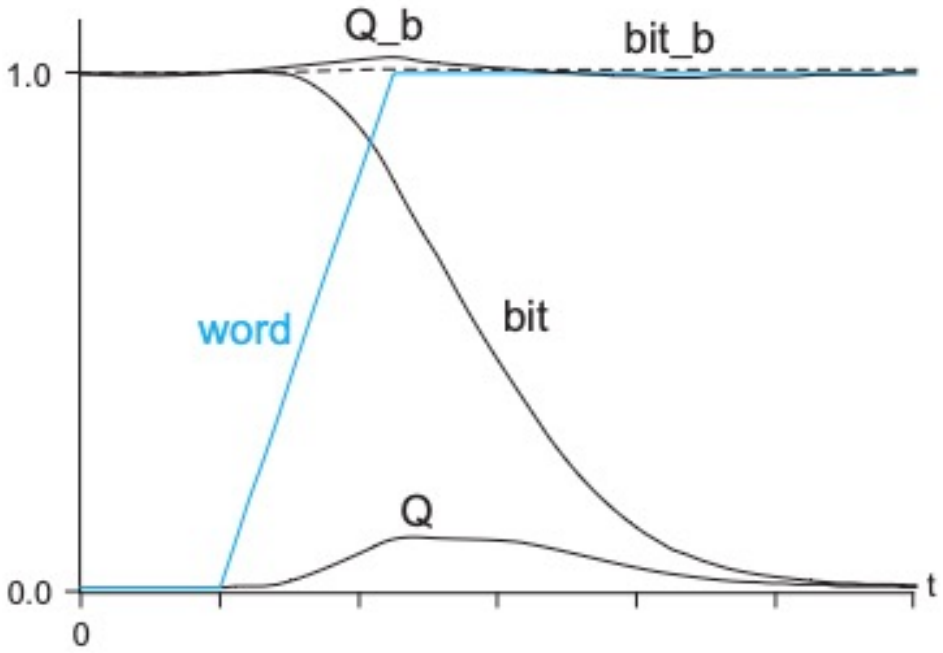




W&H p499-500

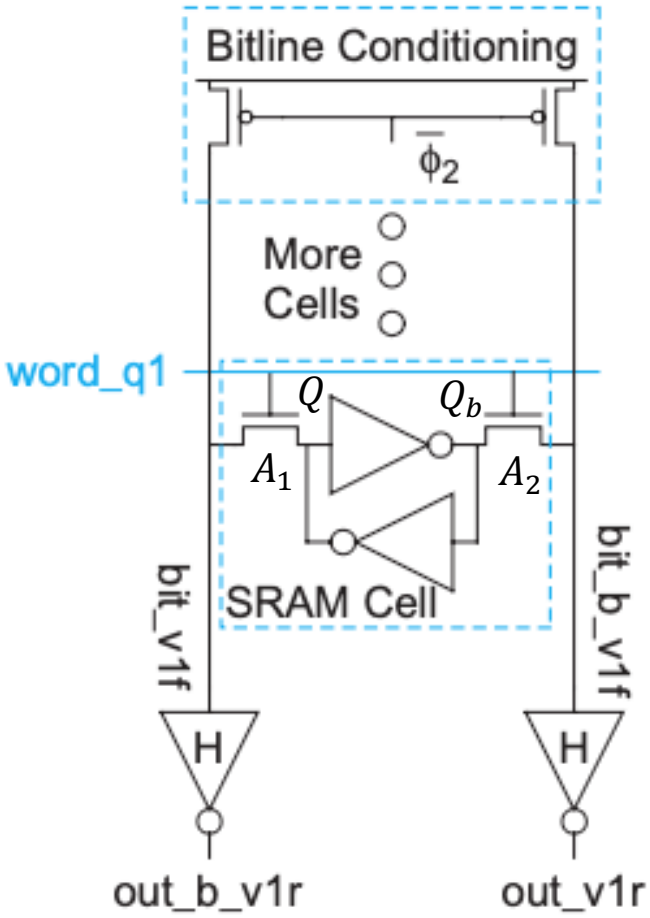
Read operation for SRAM cell

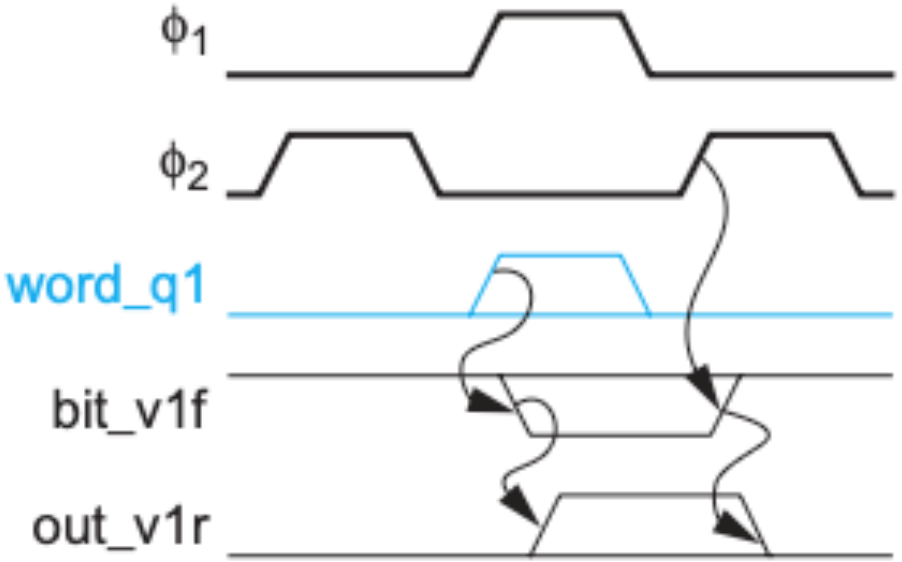




W&H p500

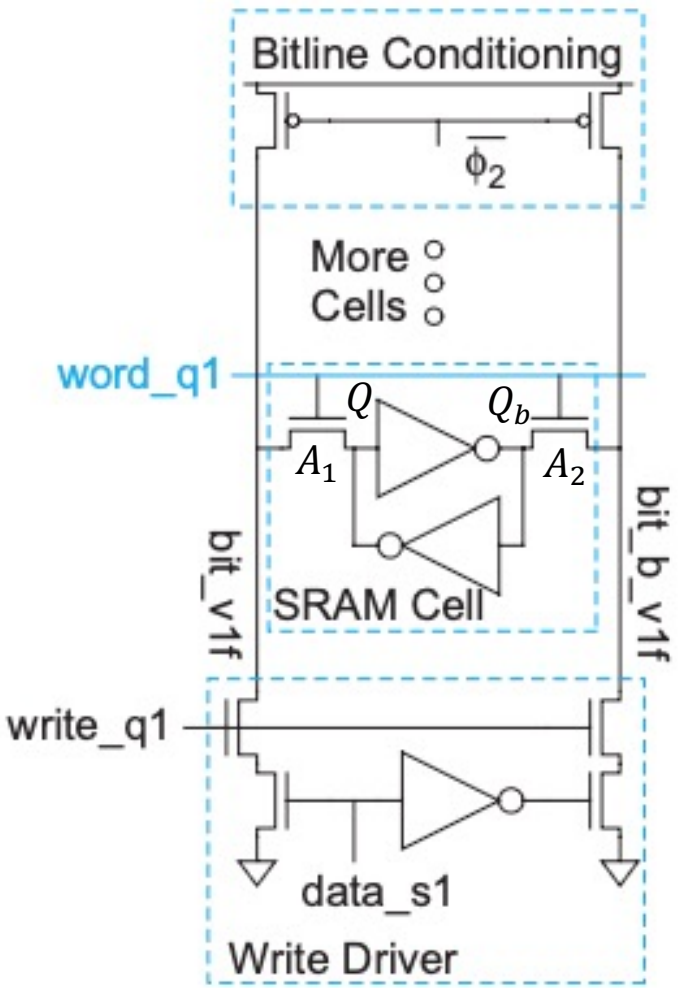
Column Read Operation

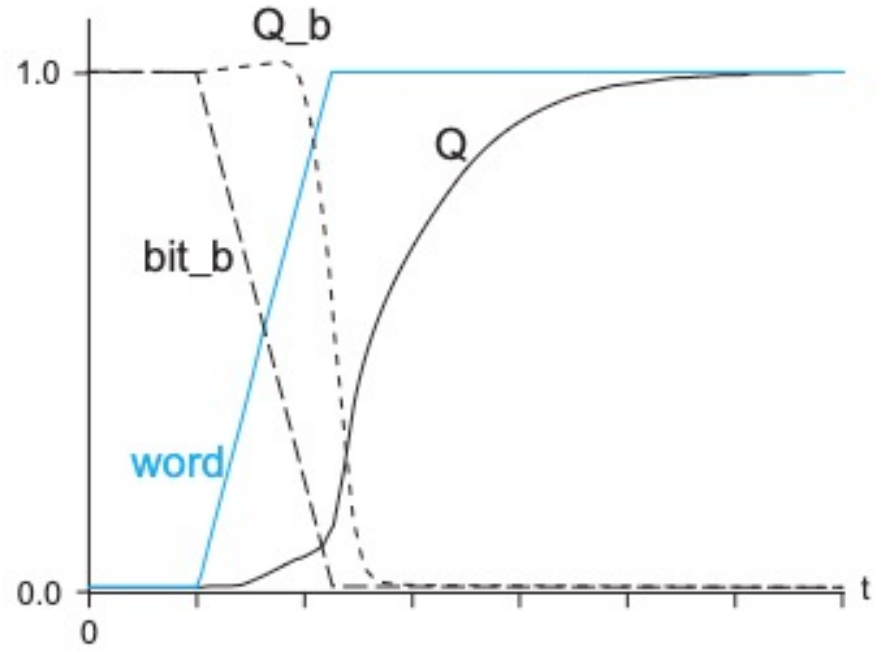


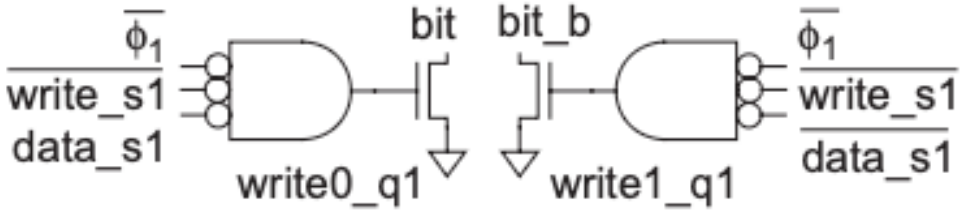


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Write Operation

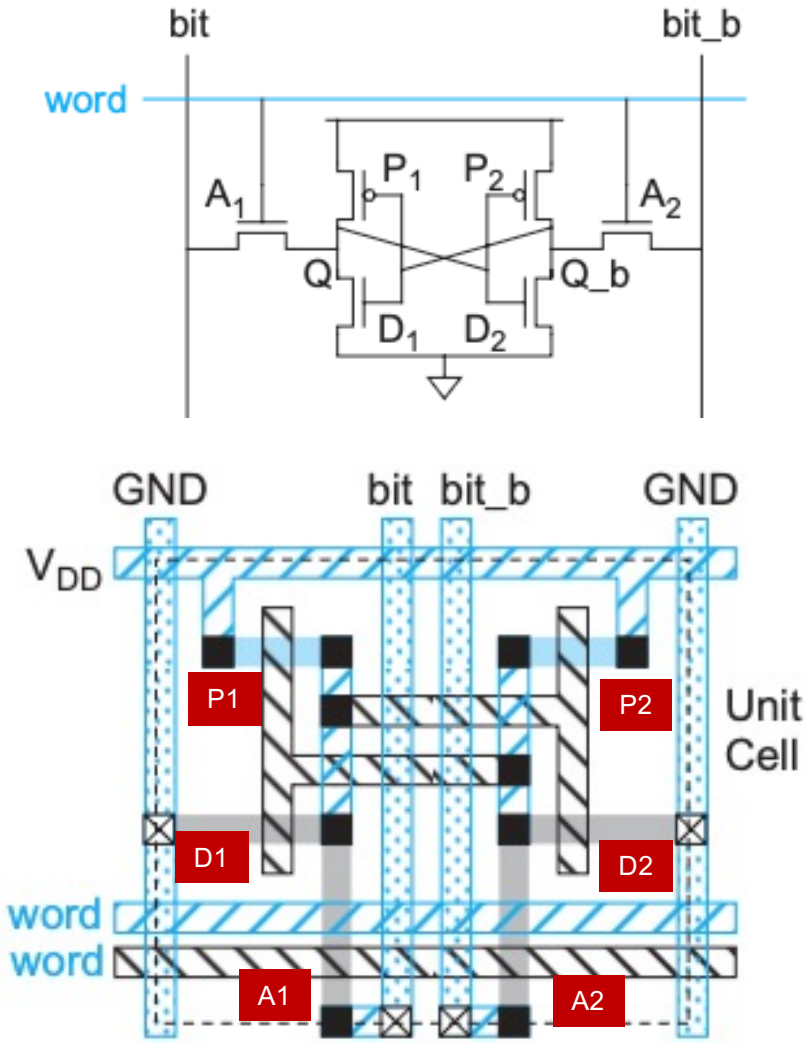


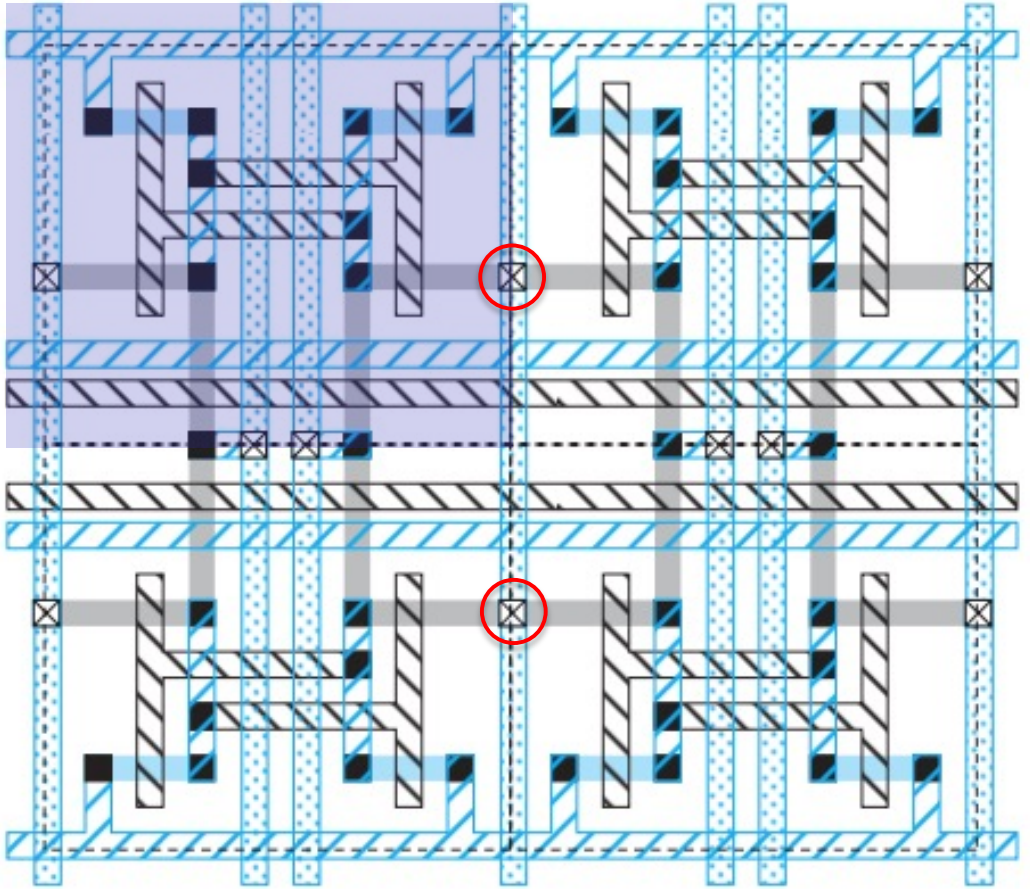




W&H p501

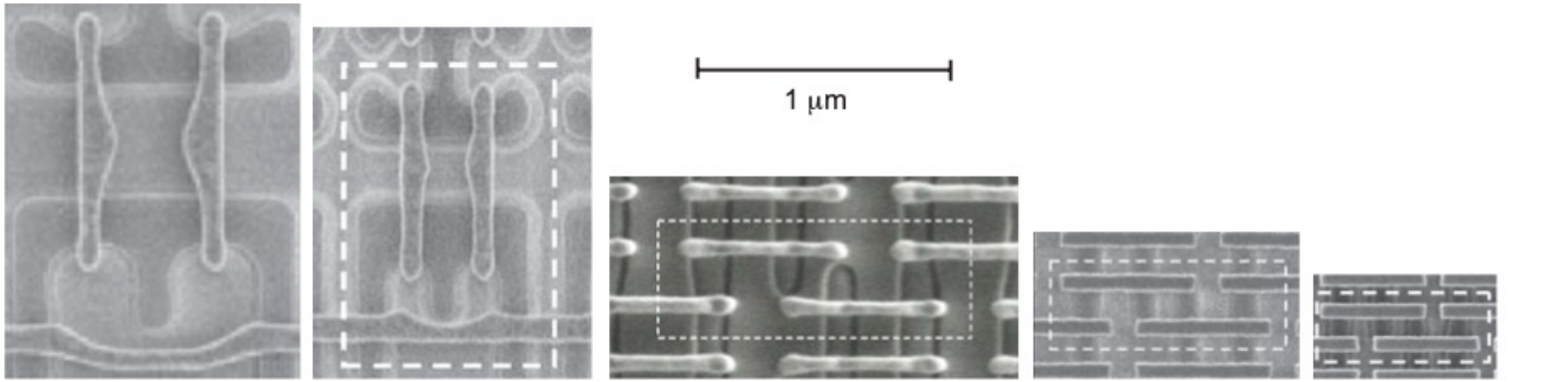
6T SRAM cell Layout (stick diagram)

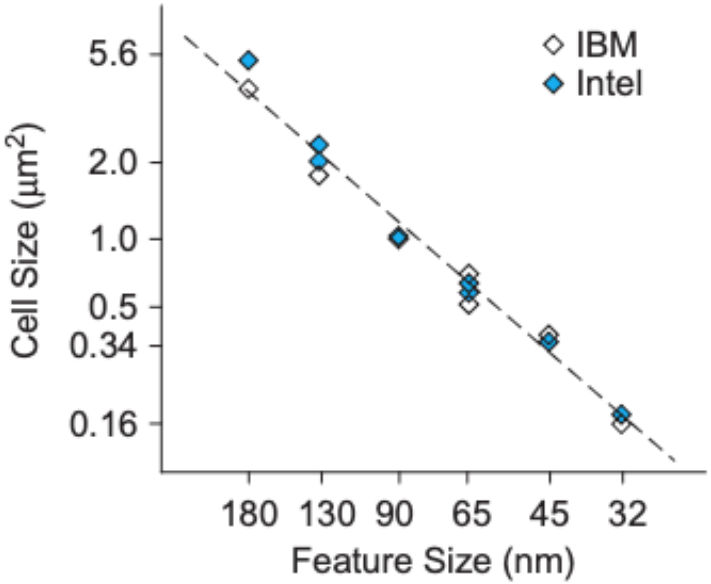




W&H p504

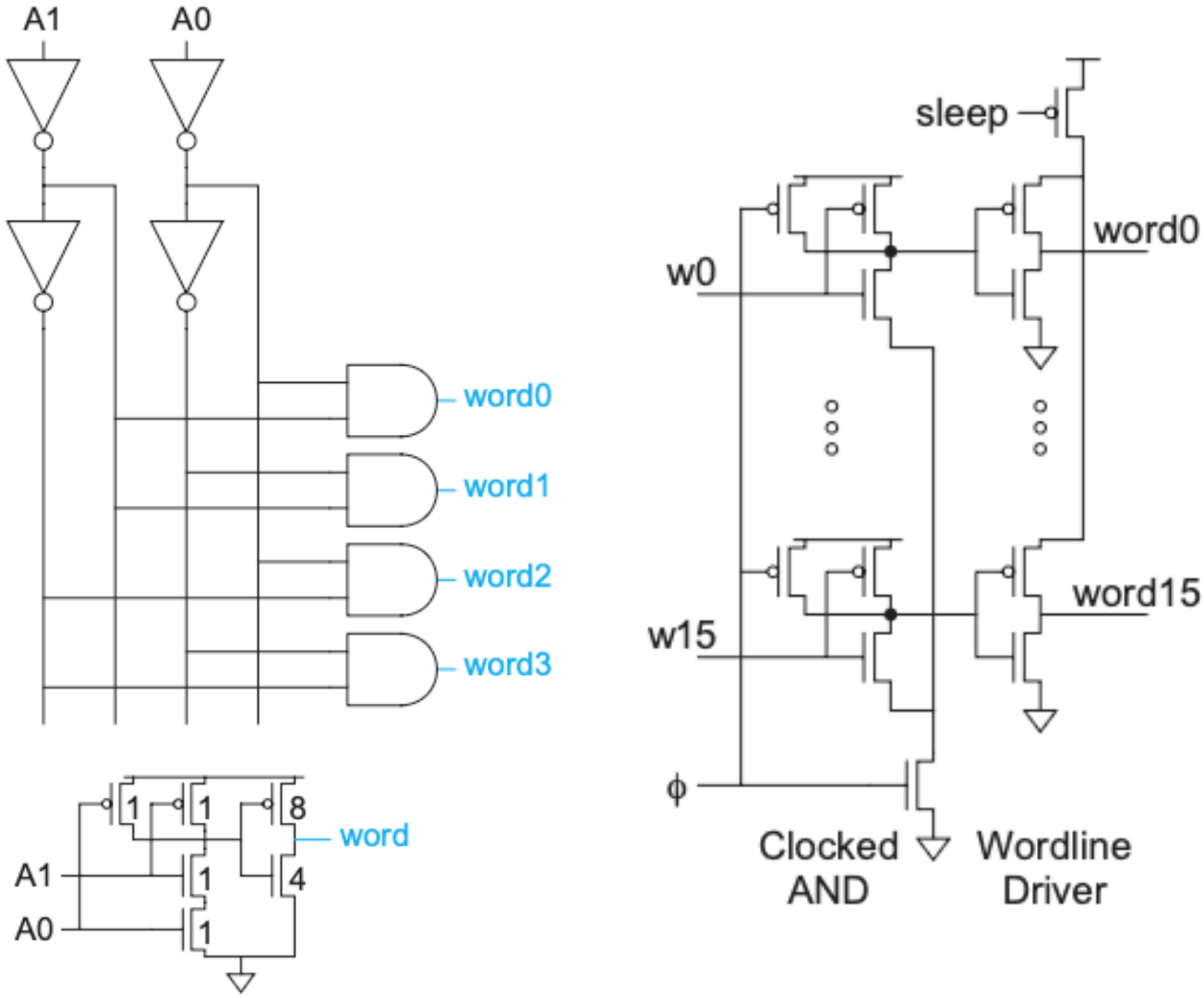
SRAM Scaling





W&H p505

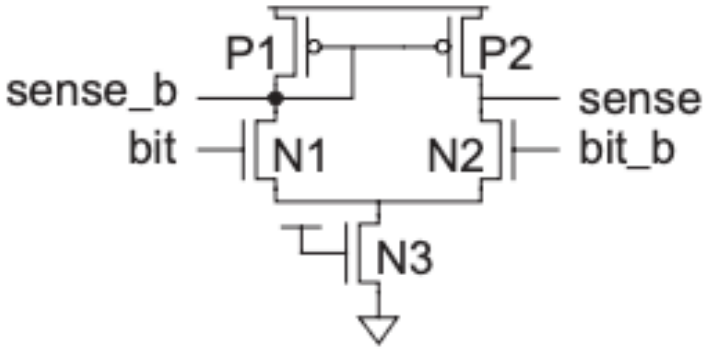
Row Decoder Circuits



W&H p506-507

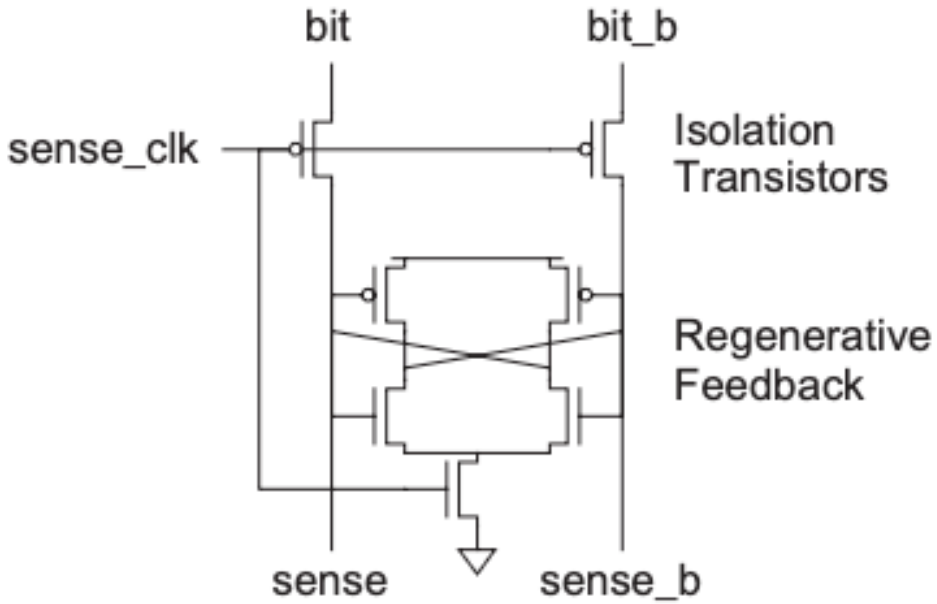
Sense Amplifier

Differntial Sense Amplifier



(a)

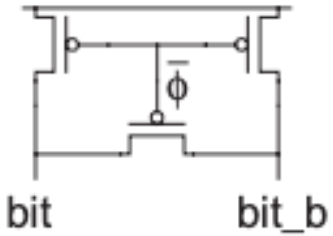
Clocked Sense Amplifier



(b)

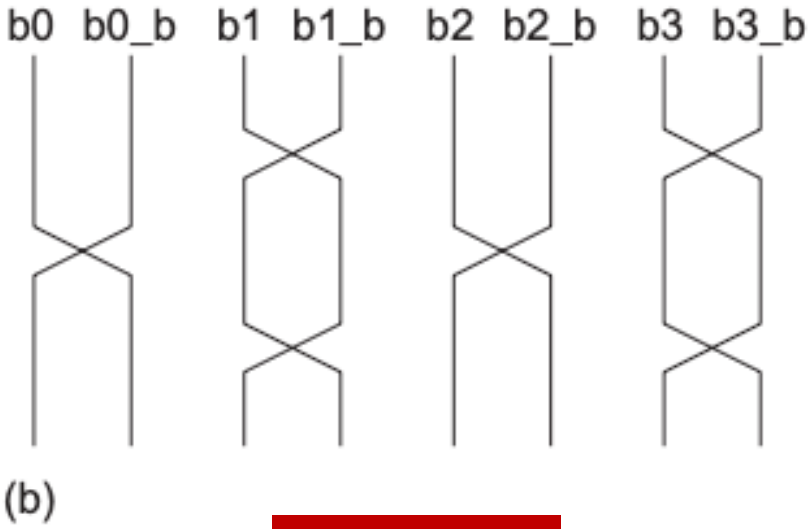
W&H p511

Differential Noise Reduction



(a)

Bitlines equalization

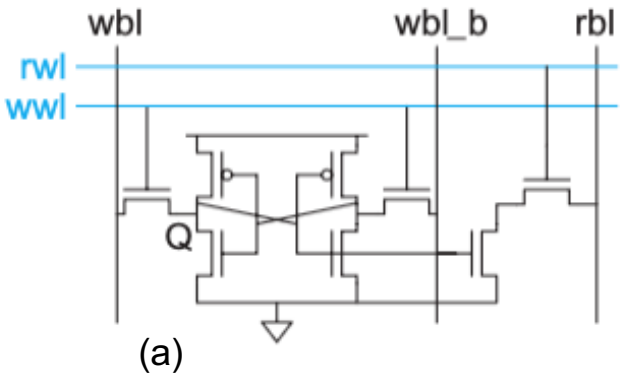


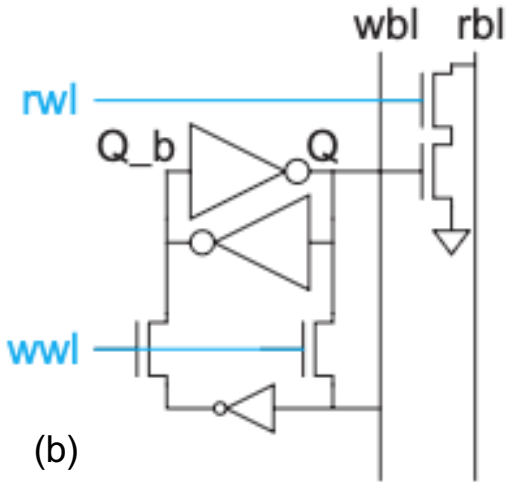
(b)

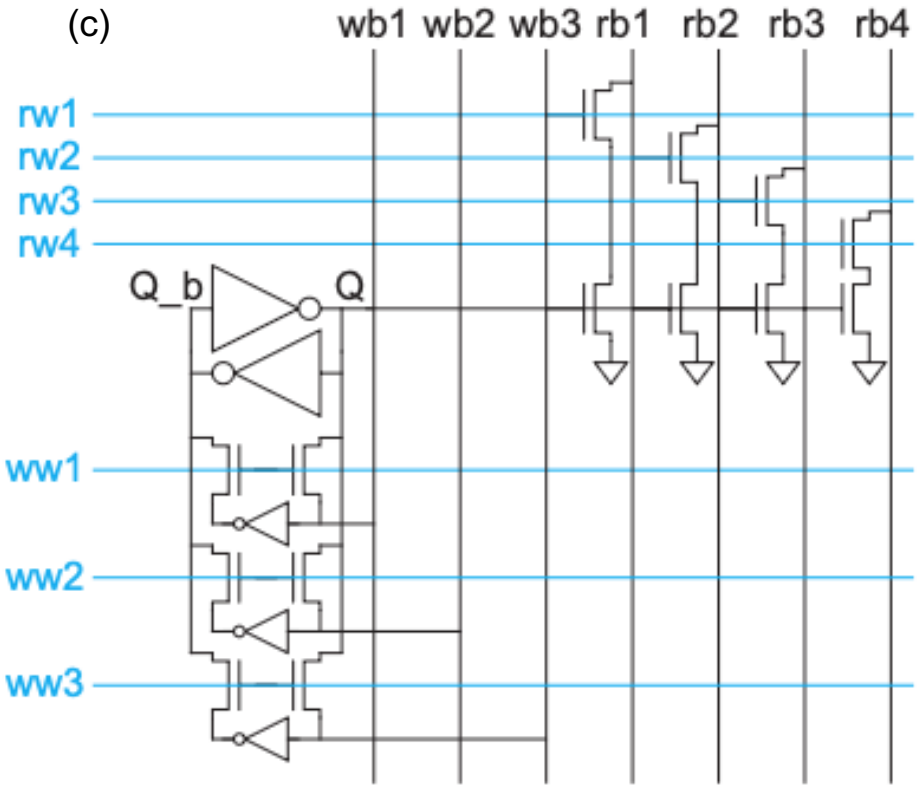
Bitlines twisting

W&H p513

Multiport SRAM



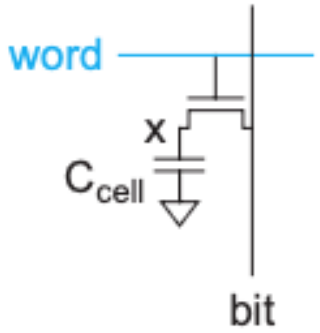


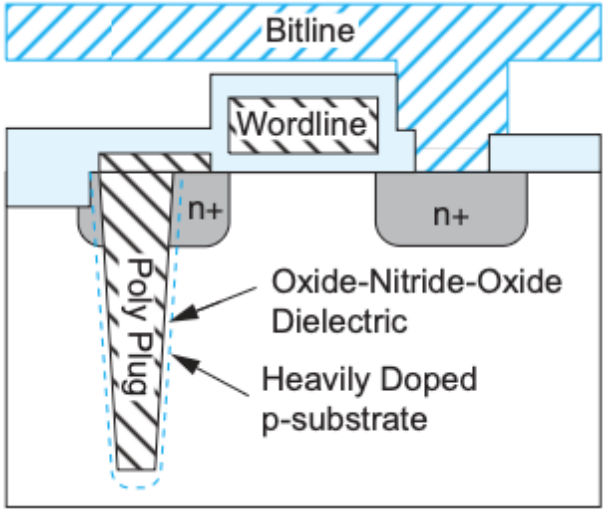


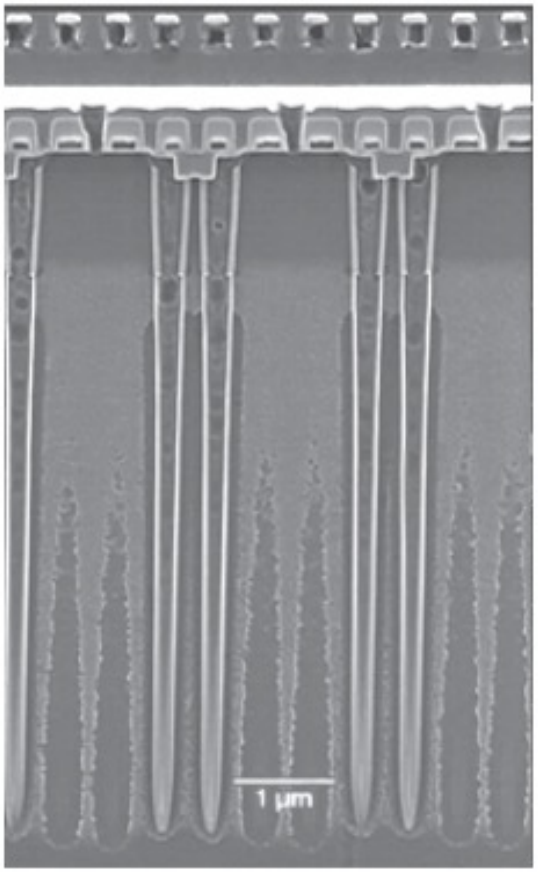
W&H p514-515

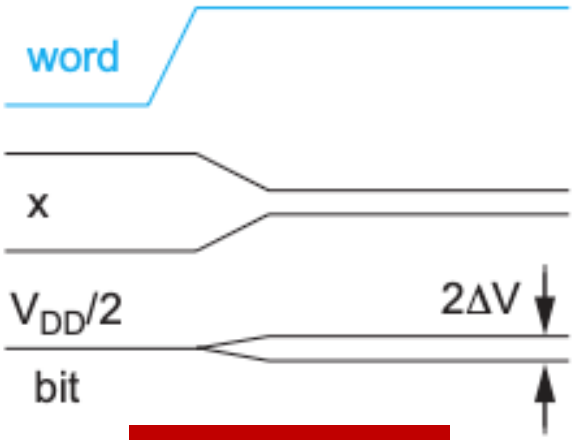
DRAM cell

Trench capacitor









VDD/2 precharge

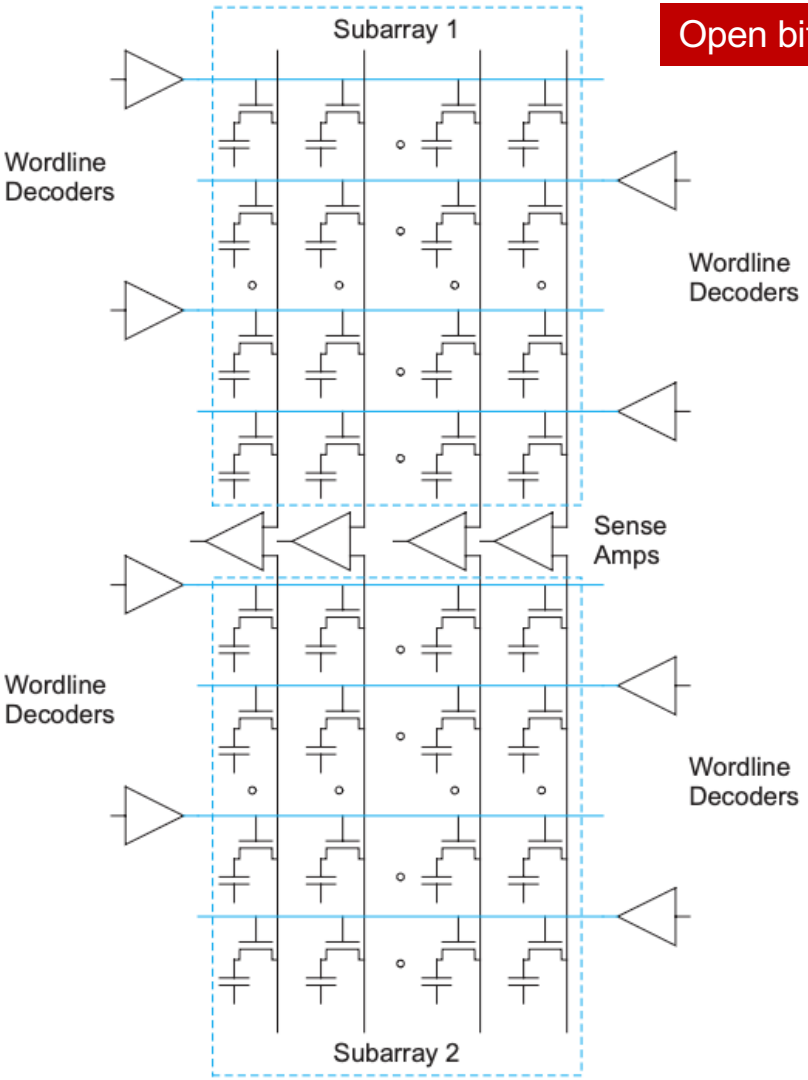
$$\Delta V = \frac{V_{DD}}{2}\frac{C_{cell}}{C_{cell}+C_{bit}}$$

Charge sharing

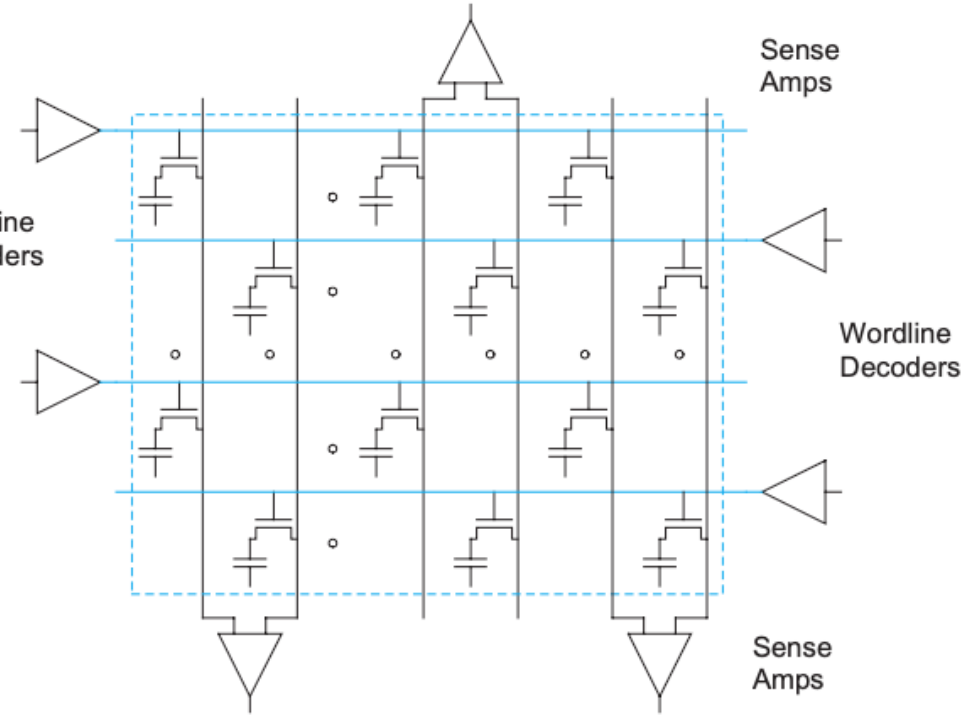
W&H p522-523

DRAM array

Open bitline



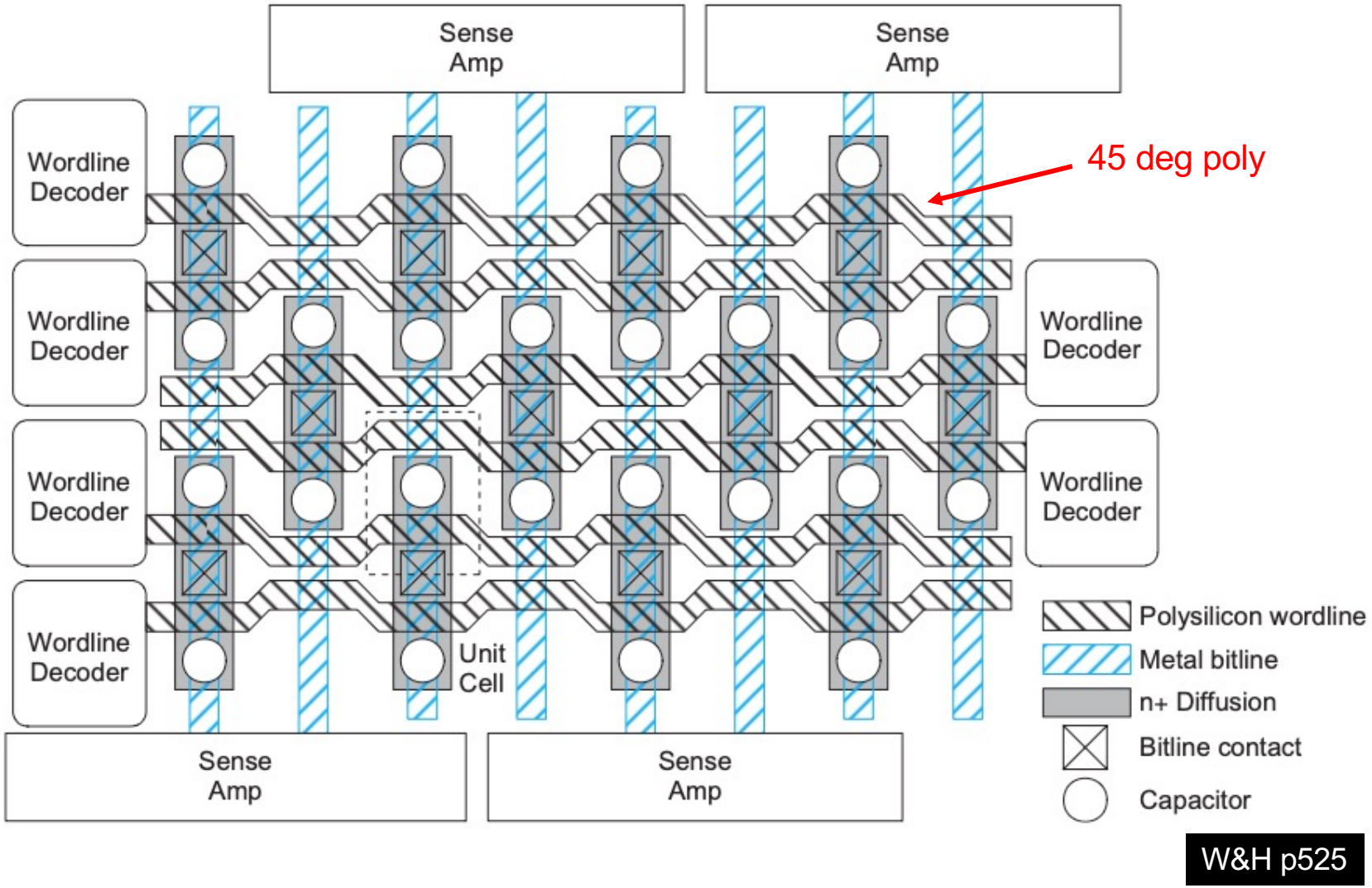
Folded bitline



- Neighbouring bitline provide reference.
- Reduce noise sensitibity

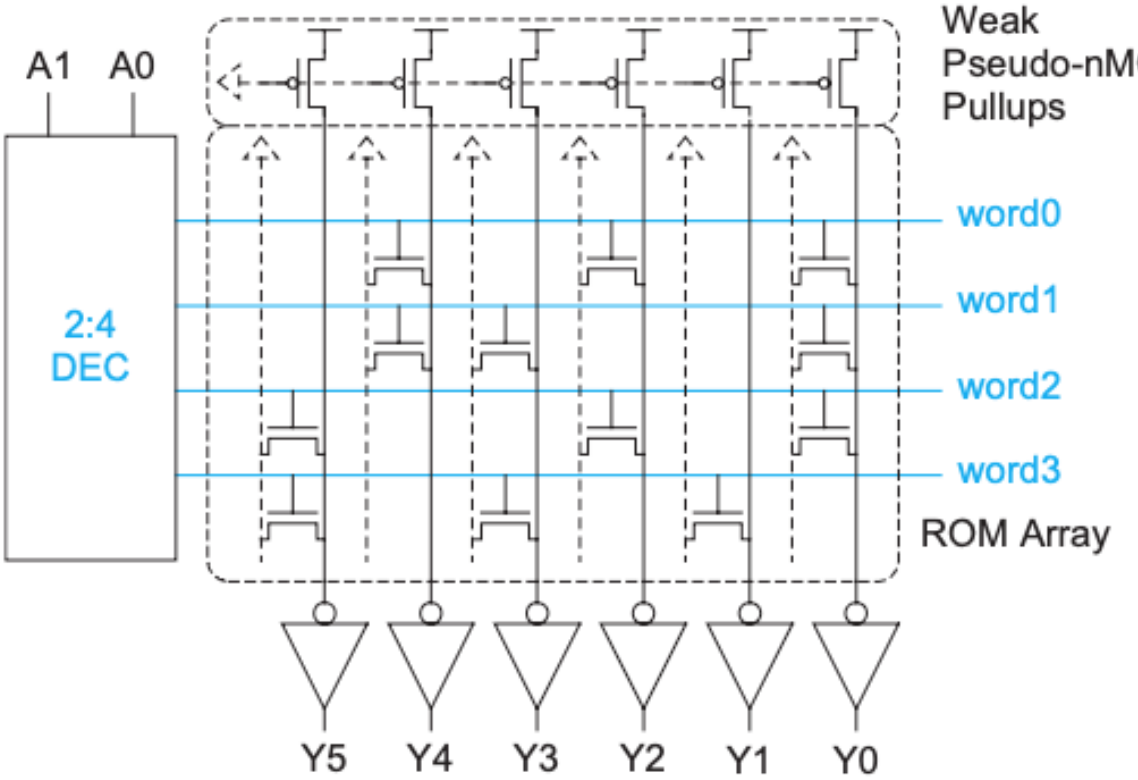
W&H p524

Folded Bitline Layout

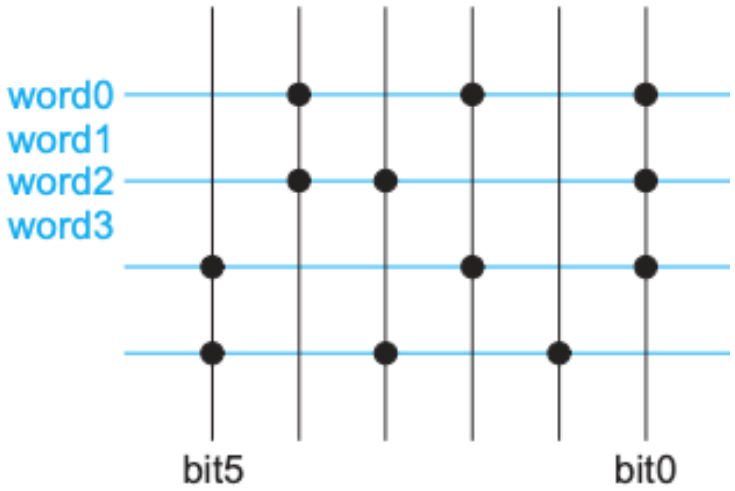


W&H p525

Pseudo nMOS ROM Circuit



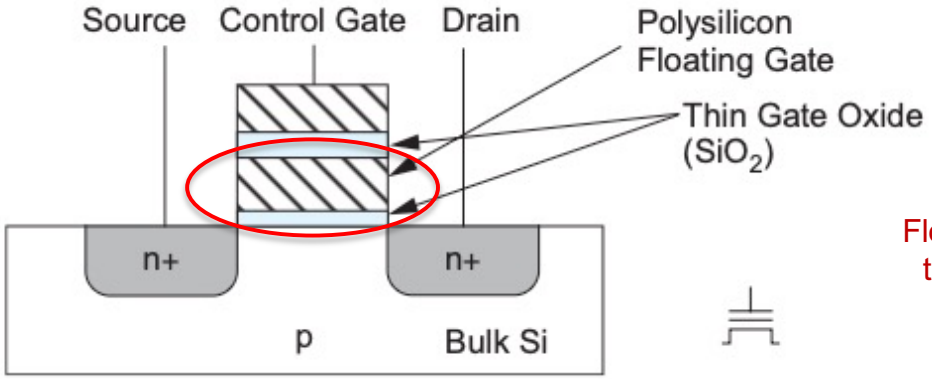

```
word0: 010101
word1: 011001
word2: 100101
word3: 101010
```



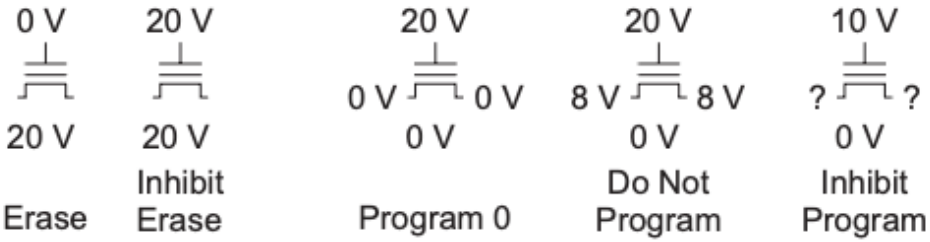
4-word by 6-bit ROM

W&H p527

Flash Memory Cell



Floating gate traps electrons through Fowler-Nordheim tunnelling

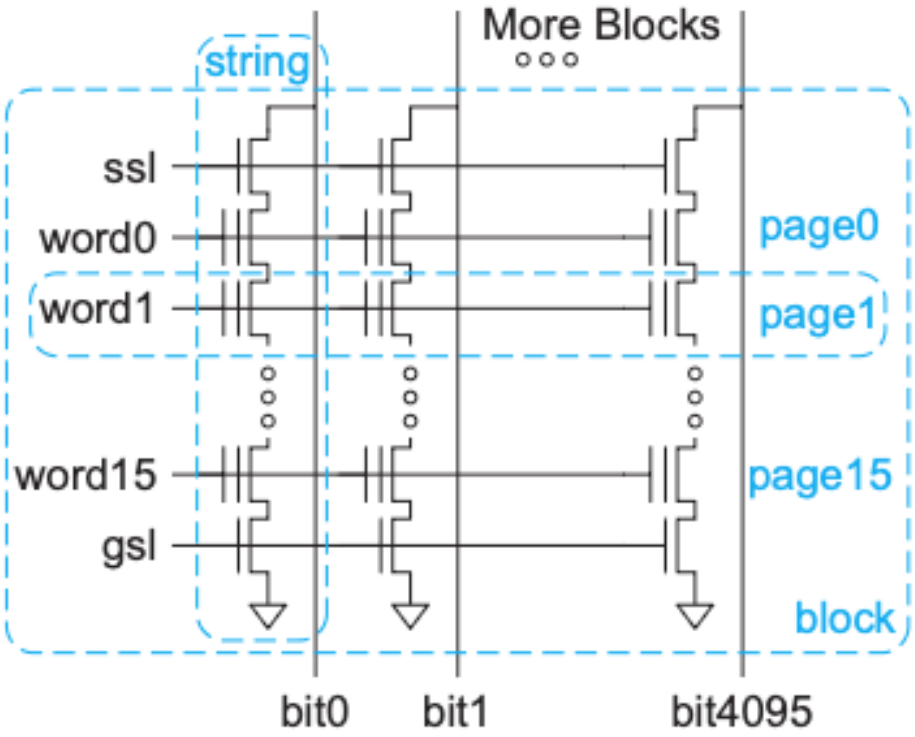


Substrate raised to 20V – cells programmed 1

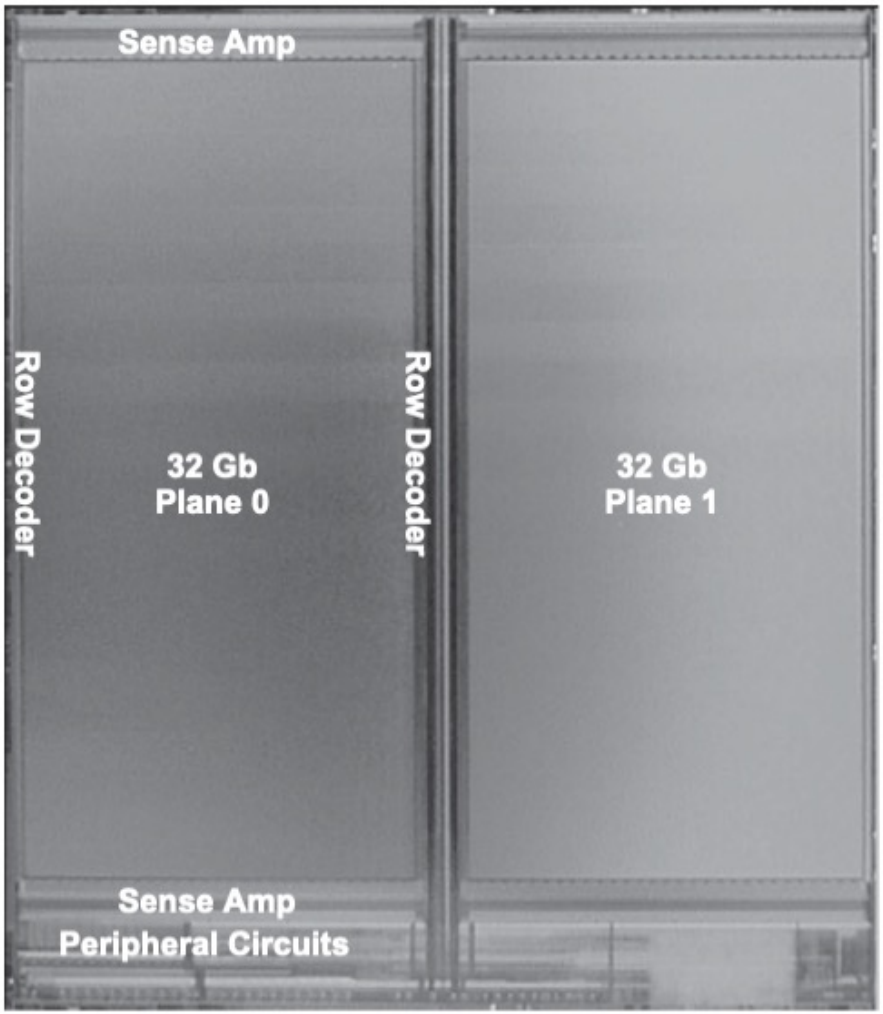
Gate raised to 20V - trap electrons raises V_t

W&H p531

Flash Memory Array

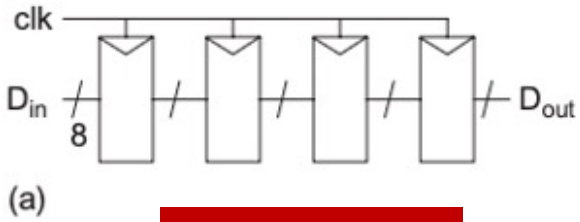


Each cell can store multibit value!

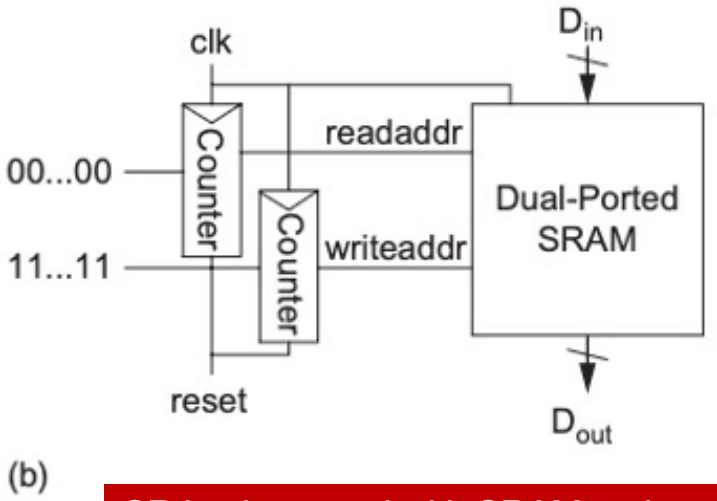


W&H p531

Shift Registers

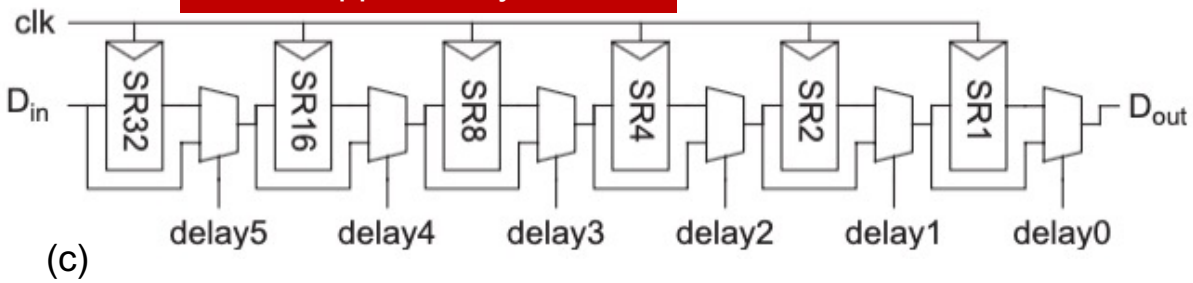


8-bit 4-stage SR



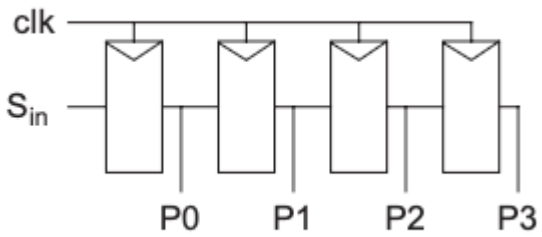
SR implemented with SRAM and counters

SR as tapped delay for DSP

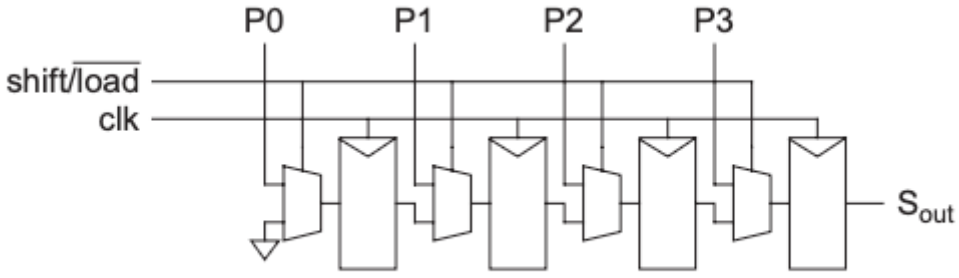


W&H p534

Delay Line and FIFO

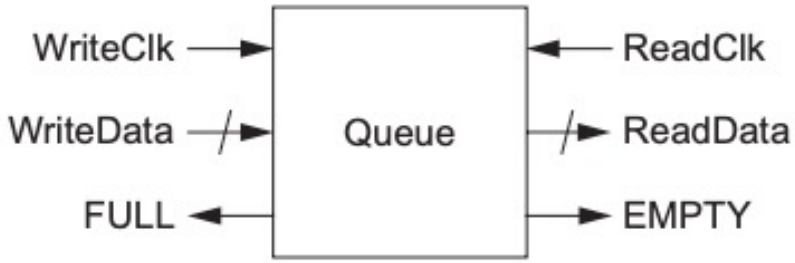


SIPO memory



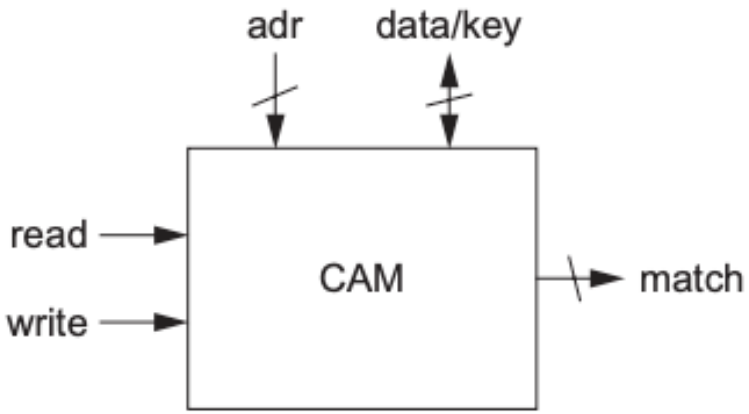
PISO memory

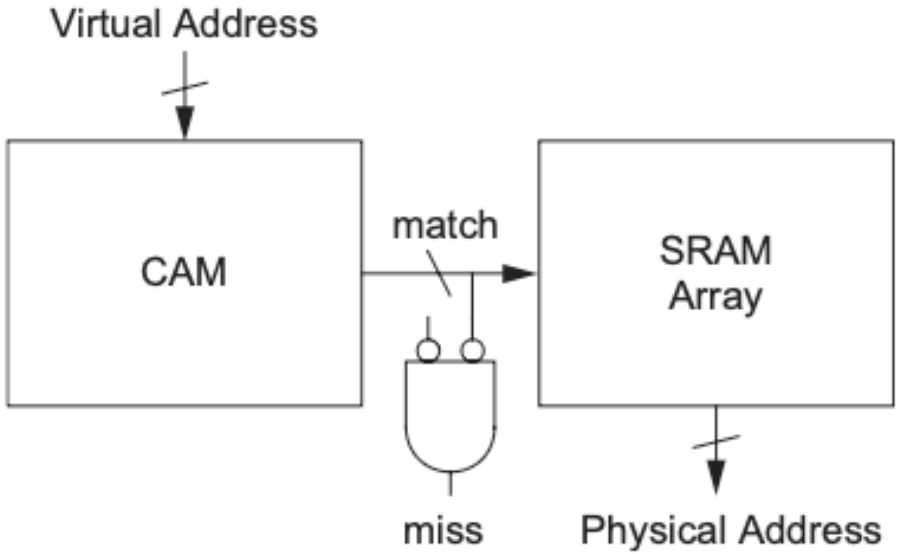
FIFO buffer



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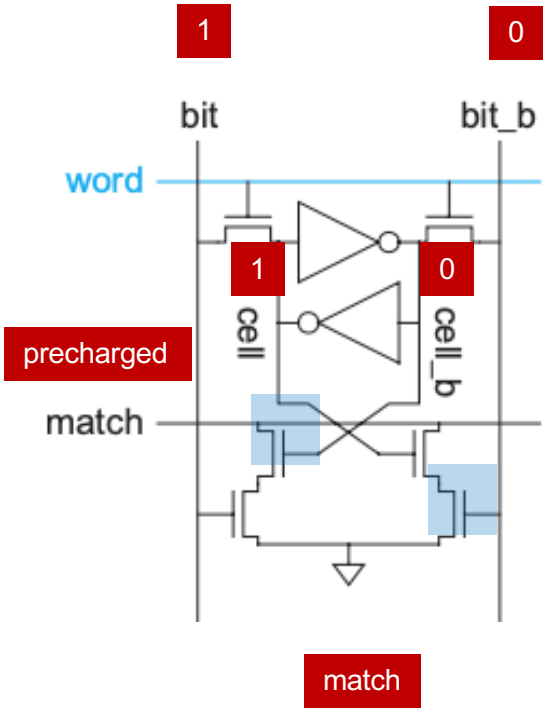
Content Addressable Memory (CAM)

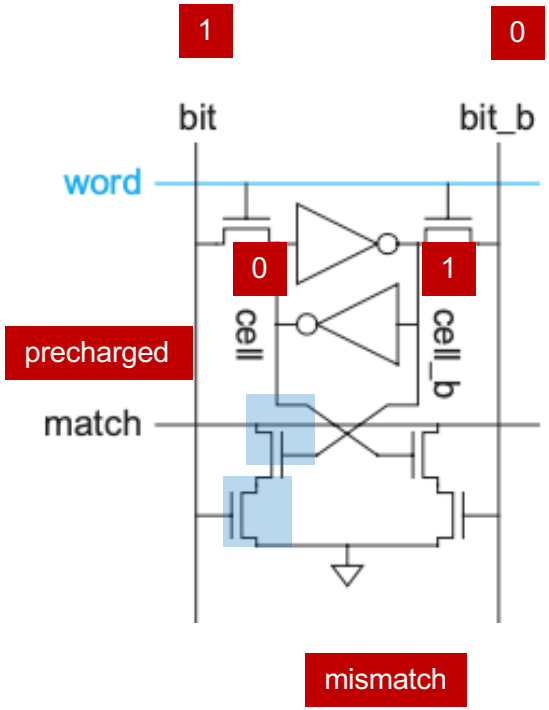




W&H p535

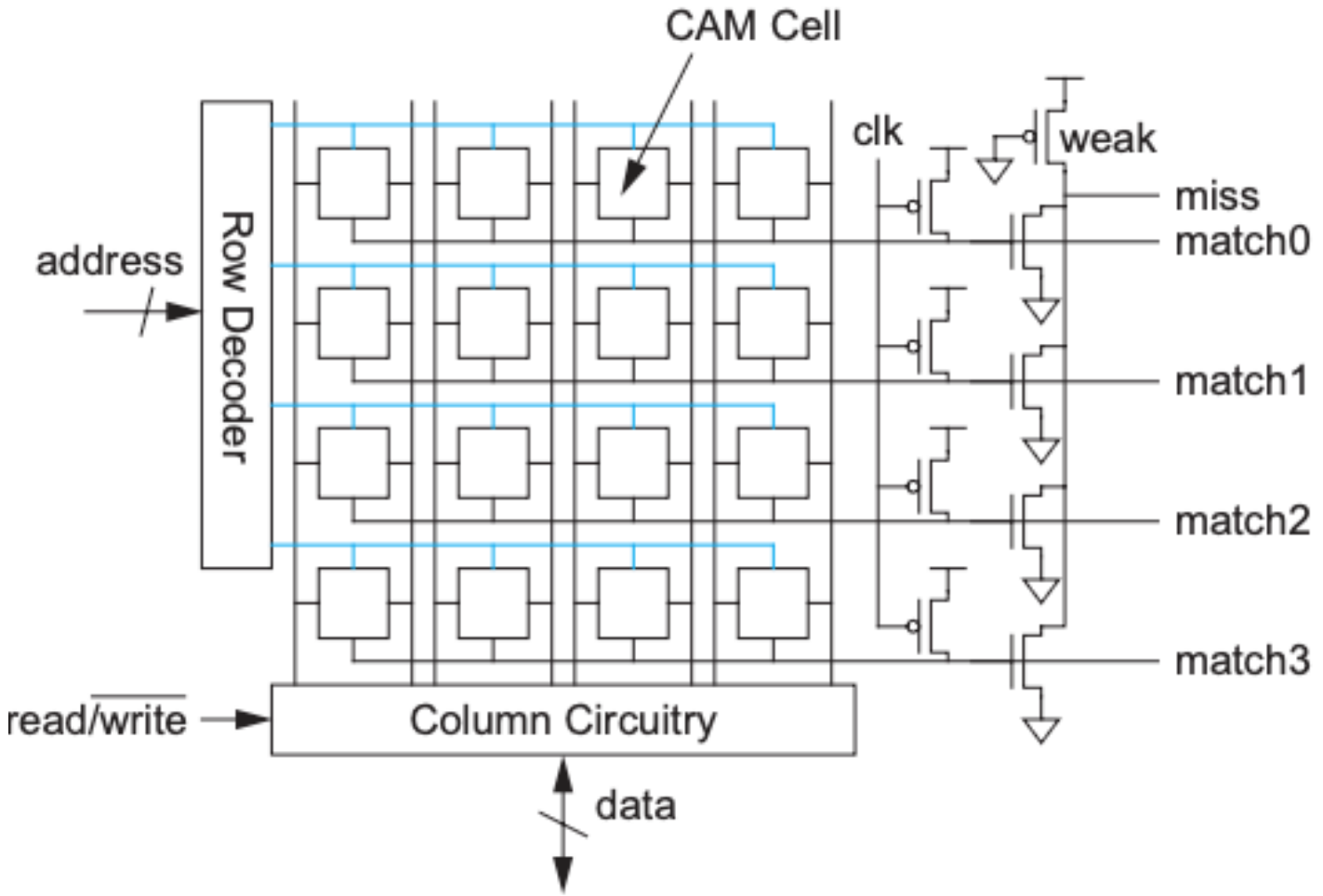
CAM Circuit (1)





W&H p535

CAM Circuit (2)



W&H p535